

ENGINEERING REVIEW / 2026-09-26

Smallest FPGA board

Components and every pin

XC7A100T, CSG324 / 15 × 15 mm package

Current result: a 33 × 36 mm board with 130 physical components, 1,139 track segments, 164 vias and four filled ground zones. All 25 boot/clock/programming nets and three power-good nets (two sequencing nets and one unconsumed status net) are connected in the native checks. The outline remains the smallest published version.

It is still electrically incomplete. There are 146 remaining power/ground connections, including seven BGA ground groups and the U4 pin 11 VSEL ground strap. All 120 mezzanine signal contacts remain unassigned. Zero physical DRC violations and zero schematic-parity issues do not establish working hardware or fabrication readiness.

This report explains every retained component and lists every numbered electrical endpoint. It replaces the earlier placement-only report for current review, while preserving that dated report as historical evidence. It focuses only on the current smallest board.

Current board	Verified result	Interpretation
Outline / layers	33 × 36 mm / 6	39 front and 91 back components; no assembly or stackup qualification claimed.
Components / endpoints	130 / 762	Every component and distinct numbered endpoint appears in the appendices.
Tracks / vias / zones	1,139 / 164 / 4	Real copper is present. Global power/ground routing remains unfinished.
Physical DRC / parity	0 / 0	Native checks, unchanged rules, no exclusions or ignored checks.
Remaining connections	146	Seven power/ground nets; not a percentage-complete metric.
ASIC interfaces	120 unassigned contacts	117-signal allocation remains a proposal; three contacts are reserved.

Read the status column carefully. “C” means the entire named net is connected in native DRC. “P” means the net still has at least one open connection; it does not assert that this particular pad is disconnected. “U” is unassigned. “NC” is an intentional no-connect.

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1 The current small board

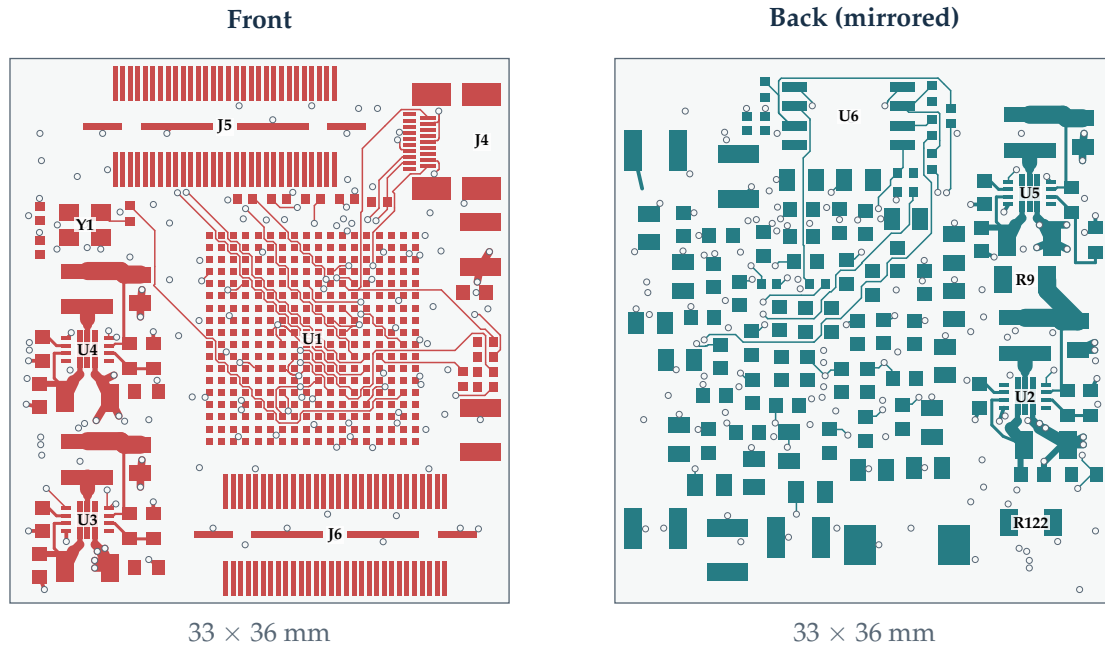


Figure 1. Vector map generated from the frozen native PCB readback. Front and mirrored back show outer-layer tracks, numbered pads and through-vias. Pad outlines are simplified to their rotated rectangular envelopes. Inner-layer tracks and ground fills are omitted here for legibility; the four native ground zones remain in the PCB. This is not an assembly drawing.

The board retains both 60-contact mezzanine connectors, the FPGA, four converters, configuration flash, oscillator and required support parts. The rail revision adds C92 and R122 on the back. Relative to the accepted compact placement, C30 moves 0.15 mm to clear routing; the remaining original component positions are preserved. There are no optional debug headers, user LEDs or standalone test-point parts.

J4's drawn body overhang remains approximately 0.65 mm, giving a board-plus-drawn-body envelope of about 33.65×36 mm. Connector mating height, full cable plug envelope, retention and dual-sided assembly clearance remain open. Unoccupied drawing area may still be needed for rail distribution, return paths, BGA escape and thermal copper.

2 How the intended circuit works

2.1 Power rails and isolation

J4.19 is the custom LINK_12V input. It is not ordinary HDMI power. The XEM8310 is the downstream FPGA module; a protected carrier power branch, cable and receiver interface still require qualification. The current headboard has no local TVS/eFuse or completed input-transient protection.

Domain	Current circuit	Purpose / boundary
Core / BRAM	U2-L1-VCORE_REG_1V025-R9-VCCINT_1V0	R9 = 15 milliohm isolates distributed bypass. VOS/feedback sense before R9.
AUX / configuration	U3-L2-VAUX_REG_1V803-R122-VCCAUX_1V8	AUX, configuration, bank14, flash, oscillator and pull-ups share 1.8 V. C92 adds bank14 bulk.
Link bank16	U4-L3-VCC_LINK_2V5	R5 = 180 kilohm, 0.1 percent. The 2.5 V rail supports the proposed link-bank direction.
ASIC-facing banks	U5-L4-VCC_ASIC_1V5	Supplies FPGA I/O banks. It does not establish safe ASIC pin voltages or external ASIC power.
Enable chain	PG_CORE enables U3; PG_AUX enables U4/U5.	Both sequencing nets and the unconsumed PGOOD_IO status net are connected. Cold-start and power-off waveforms remain unmeasured.

R12 pulls up the unconsumed PGOOD_IO status net. It is nonessential pending a remove-or-monitor decision; it does not enable a rail.

The converter-local routes include SW-to-inductor, inductor-to-output-capacitor, local input loops, quiet output-capacitor-positive sense pickup, feedback and soft-start connections. R9 and R122 must not be bypassed by subsequent copper. The downstream capacitors remain on their respective isolated rails. Routing widths describe physical geometry, not an approved current rating.

Specific remaining pin: U4.11. TPS62135 pin 11 is VSEL, not the main ground pin. Its logical net is GND, but the copper ground strap is still open. Pin 3 is the high-current ground terminal. Pin 4, FB2, is intentionally unused. A narrow logic-strap trial was rejected when it violated clearance to the new PG route. This gap remains visible in the ledger and DRC.

TI's DCS-Control example uses isolation for large distributed capacitance; local Cout/VOS remains on the regulator side. PG and discharge behavior depend on VIN. Those source requirements support this topology but do not prove stability or reverse sequencing after abrupt unplug [1]. AMD's recommended sequencing and rail-ramp limits must be checked at the FPGA, including voltage loss through the isolation elements and copper [2].

2.2 Flash, clock and JTAG

U6 is the 1.8 V MX25U12835FM2I-10G configuration flash. R103 connects FPGA CCLK to flash SCLK; R104–R107 serve DQ0–DQ3; U1.L13 drives chip select. U6.7 is RESET/SIO3 for this part. R111–R113 select M[2:0] = 001; R108–R110 pull up PROGRAM_B, INIT_B and DONE; R114 pulls PUDC_B high. CFGBVS, U1.P8, is grounded for the 1.8 V configuration domain [3,4].

Y1 is the 1.8 V ASE3-32.000MHZ-L-C-T. Its output passes through R119 to U1.P17 as the 32 MHz reference. JTAG uses dedicated FPGA pins through J4.2/15/17/18; R118 is in the TDO path. J4.1 is now the 1.8 V target-reference assignment. Programming compatibility, flash commands, clock integrity and actual cold boot remain to be demonstrated [3–5].

3 Status and remaining work

Unfinished net	Native open items	What remains
GND	8	U4.11 plus seven BGA ground groups.
LINK_12V	9	Global distribution / remaining endpoint connections.
VAUX_REG_1V803	1	Global distribution / remaining endpoint connections.
VCCAUX_1V8	49	Global distribution / remaining endpoint connections.
VCCINT_1V0	34	Global distribution / remaining endpoint connections.
VCC_ASIC_1V5	37	Global distribution / remaining endpoint connections.
VCC_LINK_2V5	8	Global distribution / remaining endpoint connections.

Do not read 146 as the full application-I/O workload. KiCad does not count an unassigned contact as a routed-net open. There are also 331 unassigned endpoints: 203 FPGA user-I/O balls, eight reserved J4 cable contacts and 120 mezzanine contacts. The six deliberate no-connects are U1.L9/L10 (DXN/DXP) and U2–U5 pin 4 (FB2).

The native schematic has 21 pages and matches the numbered PCB pad assignments. ERC still reports 331 open/unassigned pin issues and five pin-type warnings. Firmware, signal integrity, mounted power integrity, thermal performance and hardware operation have not been validated.

Decision / evidence needed	Responsible work
117 ASIC signal contract	ASIC owner / Gerald: pad type, absolute levels, direction, reset state and timing; AC_IN and IMP_TST remain especially unresolved.
Ball and connector assignment	FPGA engineering: approved 117-signal contact/ball/bank/IOSTANDARD map and matching XDC.
Downstream data link	FPGA + XEM8310 carrier engineering: protocol, receiver bias/termination, clocking, common-mode/ground-offset and JTAG voltage margins.
Global power and protection	PCB + carrier engineering: complete known rail/ground routing; qualify protected source, cable, startup, shutdown and transients.
PDN and current budget	FPGA + PCB engineering: application power estimate, actual copper/via resistance, stability, load steps and temperatures.
Mechanical / manufacture	PCB + fabricator/assembler: stackup, BGA process, connector mating, overhang, assembly tolerances and complete order codes.
Functional acceptance	Lab + FPGA engineering: cold boot, programming, reset/brownout recovery, all-channel capture, sustained transfer and error-rate tests.

4 The 117-signal interface budget

The supplied team correction adds four SPI clocks to the earlier 113-pin count. The result is 88 ASIC output/interface lines, 12 SPI L/R data streams, nine shared controls, four board clocks and four SPI clocks: **117 signals**. The 88-line group comprises eight sets of 11, including data plus associated timing/handshake lines; it is not 88 independent sampled data bits.

Two 60-contact mezzanine connectors provide 120 signal contacts, so the proposal uses 117 plus three spare contacts. Their ground-blade lands are additional to the numbered signal contacts. The proposal does not establish a completed FPGA ball map. Appendix C deliberately shows actual PCB state beside the unimplemented proposal.

Coverage. The report contains 130 component records and 762 distinct numbered endpoints: 425 assigned, 331 unassigned and six NC. The PCB has 771 numbered physical pads because duplicate shell/ground lands share endpoint numbers, plus 40 unnumbered paste/locating-hole records, for 811 total. All 48 assigned functional nets are expanded in Appendix E.

A Every component: purpose and pin connections

Each of the 130 physical parts appears once. All 120 two-terminal parts have both numbered endpoints listed here; larger devices have complete endpoint tables in Appendices B–D. Values and order codes are the saved design values, not a released purchasing BOM. “TBD” is retained wherever procurement selection is incomplete.

Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
C1 47u 25V X5R Front	Cable-input bulk charge reservoir between LINK_12V and ground. Does not replace input protection. Order: MSAST32MAB5476MPNDT1	C1.1: LINK_12V [P] C1.2: GND [P]
C2 100n 100V X7R Front	Cable-input local high-frequency bypass between LINK_12V and ground. Does not replace input protection. Order: GRM188R72A104KA35D	C2.1: LINK_12V [P] C2.2: GND [P]
C3 22u 25V X5R Back	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	C3.1: LINK_12V [P] C3.2: GND [P]
C4 100n 100V X7R Back	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	C4.1: LINK_12V [P] C4.2: GND [P]
C5 22u 16V X5R Back	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	C5.1: VCORE_REG_1V025 [C] C5.2: GND [P]
C6 10n 50V X7R Back	Soft-start capacitor from SS_CORE to ground; sets the converter reference ramp. Order: TBD-manufacturer/ordercodenotselected	C6.1: SS_CORE [C] C6.2: GND [P]
C7 22u 25V X5R Front	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	C7.1: LINK_12V [P] C7.2: GND [P]
C8 100n 100V X7R Front	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	C8.1: LINK_12V [P] C8.2: GND [P]
C9 22u 16V X5R Front	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	C9.1: VAUX_REG_1V803 [P] C9.2: GND [P]
C10 10n 50V X7R Front	Soft-start capacitor from SS_AUX to ground; sets the converter reference ramp. Order: TBD-manufacturer/ordercodenotselected	C10.1: SS_AUX [C] C10.2: GND [P]
C11 22u 25V X5R Front	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	C11.1: LINK_12V [P] C11.2: GND [P]
C12 100n 100V X7R Front	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	C12.1: LINK_12V [P] C12.2: GND [P]
C13 22u 16V X5R Front	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	C13.1: VCC_LINK_2V5 [P] C13.2: GND [P]
C14 10n 50V X7R Front	Soft-start capacitor from SS_CFG to ground; sets the converter reference ramp. Order: TBD-manufacturer/ordercodenotselected	C14.1: SS_CFG [C] C14.2: GND [P]
C15 22u 25V X5R Back	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	C15.1: LINK_12V [P] C15.2: GND [P]
C16 100n 100V X7R Back	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	C16.1: LINK_12V [P] C16.2: GND [P]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
C17 22u 16V X5R Back	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	C17.1: VCC_ASIC_1V5 [P] C17.2: GND [P]
C18 10n 50V X7R Back	Soft-start capacitor from SS_ASIC to ground; sets the converter reference ramp. Order: TBD-manufacturer/ordercodenotselected	C18.1: SS_ASIC [C] C18.2: GND [P]
C20 330u 2.5V polymer ESR25m Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: T520V337M2R5ATE025	C20.1: VCCINT_1V0 [P] C20.2: GND [P]
C21 100u 6.3V X5R Back	Distributed block-RAM decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER60J107ME20L	C21.1: VCCINT_1V0 [P] C21.2: GND [P]
C22 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C22.1: VCCINT_1V0 [P] C22.2: GND [P]
C23 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C23.1: VCCINT_1V0 [P] C23.2: GND [P]
C24 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C24.1: VCCINT_1V0 [P] C24.2: GND [P]
C25 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C25.1: VCCINT_1V0 [P] C25.2: GND [P]
C26 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C26.1: VCCINT_1V0 [P] C26.2: GND [P]
C27 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C27.1: VCCINT_1V0 [P] C27.2: GND [P]
C28 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C28.1: VCCINT_1V0 [P] C28.2: GND [P]
C29 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C29.1: VCCINT_1V0 [P] C29.2: GND [P]
C30 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C30.1: VCCINT_1V0 [P] C30.2: GND [P]
C31 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C31.1: VCCINT_1V0 [P] C31.2: GND [P]
C32 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C32.1: VCCINT_1V0 [P] C32.2: GND [P]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
C33 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C33.1: VCCINT_1V0 [P] C33.2: GND [P]
C34 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C34.1: VCCINT_1V0 [P] C34.2: GND [P]
C35 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C35.1: VCCINT_1V0 [P] C35.2: GND [P]
C36 470n 6.3V X7R Back	Distributed block-RAM decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C36.1: VCCINT_1V0 [P] C36.2: GND [P]
C37 470n 6.3V X7R Back	Distributed block-RAM decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C37.1: VCCINT_1V0 [P] C37.2: GND [P]
C40 47u 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER70J476ME20L	C40.1: VCCAUX_1V8 [P] C40.2: GND [P]
C41 4.7u 10V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C41.1: VCCAUX_1V8 [P] C41.2: GND [P]
C42 4.7u 10V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C42.1: VCCAUX_1V8 [P] C42.2: GND [P]
C43 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C43.1: VCCAUX_1V8 [P] C43.2: GND [P]
C44 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C44.1: VCCAUX_1V8 [P] C44.2: GND [P]
C45 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C45.1: VCCAUX_1V8 [P] C45.2: GND [P]
C46 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C46.1: VCCAUX_1V8 [P] C46.2: GND [P]
C50 47u 6.3V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM32ER70J476ME20L	C50.1: VCCAUX_1V8 [P] C50.2: GND [P]
C51 4.7u 10V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM21BR71A475KA73L	C51.1: VCCAUX_1V8 [P] C51.2: GND [P]
C52 4.7u 10V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM21BR71A475KA73L	C52.1: VCCAUX_1V8 [P] C52.2: GND [P]
C53 470n 6.3V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM188R70J474KA01D	C53.1: VCCAUX_1V8 [P] C53.2: GND [P]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
C54 470n 6.3V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM188R70J474KA01D	C54.1: VCCAUX_1V8 [P] C54.2: GND [P]
C55 470n 6.3V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM188R70J474KA01D	C55.1: VCCAUX_1V8 [P] C55.2: GND [P]
C56 470n 6.3V X7R Back	Bypass for configuration/bank14 on the shared 1.8 V AUX rail, downstream of R122. Order: GRM188R70J474KA01D	C56.1: VCCAUX_1V8 [P] C56.2: GND [P]
C57 4.7u 10V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM21BR71A475KA73L	C57.1: VCC_LINK_2V5 [P] C57.2: GND [P]
C58 4.7u 10V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM21BR71A475KA73L	C58.1: VCC_LINK_2V5 [P] C58.2: GND [P]
C59 470n 6.3V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM188R70J474KA01D	C59.1: VCC_LINK_2V5 [P] C59.2: GND [P]
C60 470n 6.3V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM188R70J474KA01D	C60.1: VCC_LINK_2V5 [P] C60.2: GND [P]
C61 470n 6.3V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM188R70J474KA01D	C61.1: VCC_LINK_2V5 [P] C61.2: GND [P]
C62 470n 6.3V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM188R70J474KA01D	C62.1: VCC_LINK_2V5 [P] C62.2: GND [P]
C63 47u 6.3V X7R Back	Local/bulk bypass for bank16 on the 2.5 V link rail; final plane connection and PDN performance remain incomplete. Order: GRM32ER70J476ME20L	C63.1: VCC_LINK_2V5 [P] C63.2: GND [P]
C64 4.7u 10V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C64.1: VCC_ASIC_1V5 [P] C64.2: GND [P]
C65 4.7u 10V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C65.1: VCC_ASIC_1V5 [P] C65.2: GND [P]
C66 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C66.1: VCC_ASIC_1V5 [P] C66.2: GND [P]
C67 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C67.1: VCC_ASIC_1V5 [P] C67.2: GND [P]
C68 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C68.1: VCC_ASIC_1V5 [P] C68.2: GND [P]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
C69 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C69.1: VCC_ASIC_1V5 [P] C69.2: GND [P]
C70 4.7u 10V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C70.1: VCC_ASIC_1V5 [P] C70.2: GND [P]
C71 4.7u 10V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C71.1: VCC_ASIC_1V5 [P] C71.2: GND [P]
C72 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C72.1: VCC_ASIC_1V5 [P] C72.2: GND [P]
C73 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C73.1: VCC_ASIC_1V5 [P] C73.2: GND [P]
C74 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C74.1: VCC_ASIC_1V5 [P] C74.2: GND [P]
C75 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C75.1: VCC_ASIC_1V5 [P] C75.2: GND [P]
C76 4.7u 10V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C76.1: VCC_ASIC_1V5 [P] C76.2: GND [P]
C77 4.7u 10V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	C77.1: VCC_ASIC_1V5 [P] C77.2: GND [P]
C78 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C78.1: VCC_ASIC_1V5 [P] C78.2: GND [P]
C79 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C79.1: VCC_ASIC_1V5 [P] C79.2: GND [P]
C80 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C80.1: VCC_ASIC_1V5 [P] C80.2: GND [P]
C81 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	C81.1: VCC_ASIC_1V5 [P] C81.2: GND [P]
C82 47u 6.3V X7R Front	Distributed ASIC-facing I/O bulk decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER70J476ME20L	C82.1: VCC_ASIC_1V5 [P] C82.2: GND [P]
C90 100n 16V X7R Back	Bypass for the FPGA analog/XADC supply domain, tied to the auxiliary rail in this design. Order: TBD-manufacturer/ordercodenotselected	C90.1: VCCAUX_1V8 [P] C90.2: GND [P]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
C91 1u 10V X7R Back	Bypass for the FPGA analog/XADC supply domain, tied to the auxiliary rail in this design. Order: TBD-manufacturer/ordercodenotselected	C91.1: VCCAUX_1V8 [P] C91.2: GND [P]
C92 47u 6.3V X7R Back	Bank14 bulk bypass on the shared 1.8 V AUX/configuration rail, downstream of R122. Order: GRM32ER70J476ME20L	C92.1: VCCAUX_1V8 [P] C92.2: GND [P]
C100 100n Back	Local flash supply decoupling from 1.8 V to ground. Order: TBD_0402_X7R_16V_100n	C100.1: VCCAUX_1V8 [P] C100.2: GND [P]
C101 1u Back	Local flash supply decoupling from 1.8 V to ground. Order: TBD_0402_X7R_16V_1u	C101.1: VCCAUX_1V8 [P] C101.2: GND [P]
C102 10n Front	Local oscillator supply decoupling from 1.8 V to ground. Order: TBD_0402_X7R_16V_10n	C102.1: VCCAUX_1V8 [P] C102.2: GND [P]
C103 100n Front	Local oscillator supply decoupling from 1.8 V to ground. Order: TBD_0402_X7R_16V_100n	C103.1: VCCAUX_1V8 [P] C103.2: GND [P]
J4 CUSTOM POWER + JTAG / DATA TBD Front	Single custom cable port for proposed 12 V input, JTAG and four reserved data pairs. It is not wired as ordinary HDMI. Order: Molex46765family;exactordercodepending	All cable pins: Appendix C.
J5 QSH-030-01-L-D-A Front	60-contact routing-board mezzanine plus four ground-blade lands and two locating holes. Proposed ASIC contact allocation is separate from CAD. Order: QSH-030-01-L-D-A	All mezzanine contacts: Appendix C.
J6 QSH-030-01-L-D-A Front	60-contact routing-board mezzanine plus four ground-blade lands and two locating holes. Proposed ASIC contact allocation is separate from CAD. Order: QSH-030-01-L-D-A	All mezzanine contacts: Appendix C.
L1 1uH 5.4A Isat30% Back	Buck energy-storage inductor between SW_CORE and VCORE_REG_1V025. Keep with its converter and local capacitors. Order: XFL4020-102MEC	L1.1: SW_CORE [C] L1.2: VCORE_REG_1V025 [C]
L2 1uH 5.4A Isat30% Front	Buck energy-storage inductor between SW_AUX and VAUX_REG_1V803, upstream of R122. Keep with its converter and local capacitors. Order: XFL4020-102MEC	L2.1: SW_AUX [C] L2.2: VAUX_REG_1V803 [P]
L3 1uH 5.4A Isat30% Front	Buck energy-storage inductor between SW_CFG and the 2.5 V VCC_LINK_2V5 output. Keep with its converter and local capacitors. Order: XFL4020-102MEC	L3.1: SW_CFG [C] L3.2: VCC_LINK_2V5 [P]
L4 1uH 5.4A Isat30% Back	Buck energy-storage inductor between SW_ASIC and VCC_ASIC_1V5. Keep with its converter and local capacitors. Order: XFL4020-102MEC	L4.1: SW_ASIC [C] L4.2: VCC_ASIC_1V5 [P]
R1 32.4k 0.1% Back	Upper feedback-divider resistor from the regulated output to FB_CORE. Sets the output together with the lower resistor. Order: TBD-manufacturer/ordercodenotselected	R1.1: VCORE_REG_1V025 [C] R1.2: FB_CORE [C]
R2 69.8k 0.1% Back	Lower feedback-divider resistor from FB_CORE to ground. Retain its ratio with the upper resistor. Order: TBD-manufacturer/ordercodenotselected	R2.1: FB_CORE [C] R2.2: GND [P]
R3 110k 0.1% Front	Upper feedback-divider resistor from the regulated output to FB_AUX. Sets the output together with the lower resistor. Order: TBD-manufacturer/ordercodenotselected	R3.1: VAUX_REG_1V803 [P] R3.2: FB_AUX [C]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
R4 69.8k 0.1% Front	Lower feedback-divider resistor from FB_AUX to ground. Retain its ratio with the upper resistor. Order: TBD-manufacturer/ordercodenotselected	R4.1: FB_AUX [C] R4.2: GND [P]
R5 180k 0.1% Front	Upper feedback-divider resistor for U4; 180 kilohm sets the nominal 2.5 V link-bank rail with R6. Order: TNPW0603180KBEEA	R5.1: VCC_LINK_2V5 [P] R5.2: FB_CFG [C]
R6 69.8k 0.1% Front	Lower feedback-divider resistor from FB_CFG to ground. Retain its ratio with the upper resistor. Order: TBD-manufacturer/ordercodenotselected	R6.1: FB_CFG [C] R6.2: GND [P]
R7 80.6k 0.1% Back	Upper feedback-divider resistor from the regulated output to FB_ASIC. Sets the output together with the lower resistor. Order: TBD-manufacturer/ordercodenotselected	R7.1: VCC_ASIC_1V5 [P] R7.2: FB_ASIC [C]
R8 69.8k 0.1% Back	Lower feedback-divider resistor from FB_ASIC to ground. Retain its ratio with the upper resistor. Order: TBD-manufacturer/ordercodenotselected	R8.1: FB_ASIC [C] R8.2: GND [P]
R9 15m 1% 1W Back	15 milliohm series element between the sensed converter output and distributed FPGA core rail. Bulk isolation causes load-dependent voltage drop. Order: WSLP1206R0150FEA	R9.1: VCORE_REG_1V025 [C] R9.2: VCCINT_1V0 [P]
R10 100k Back	Pull-up for PG_CORE to LINK_12V. Allows an open-drain power-good output to rise. Order: TBD-manufacturer/ordercodenotselected	R10.1: LINK_12V [P] R10.2: PG_CORE [C]
R11 100k Front	Pull-up for PG_AUX to LINK_12V. Allows an open-drain power-good output to rise. Order: TBD-manufacturer/ordercodenotselected	R11.1: LINK_12V [P] R11.2: PG_AUX [C]
R12 100k Front	PGOOD_IO status pull-up. The status net has no consumer; this part is nonessential pending a remove-or-monitor decision. Order: TBD-manufacturer/ordercodenotselected	R12.1: VCCAUX_1V8 [P] R12.2: PGOOD_IO [C]
R100 2.4k Back	Pull-up keeping FLASH_CS_B high when it is not driven. Supports inactive flash-control state. Order: TBD_0402_1pct_2.4k	R100.1: VCCAUX_1V8 [P] R100.2: FLASH_CS_B [C]
R101 4.7k Back	Pull-up keeping FLASH_DQ2 high when it is not driven. Supports inactive flash-control state. Order: TBD_0402_1pct_4.7k	R101.1: VCCAUX_1V8 [P] R101.2: FLASH_DQ2 [C]
R102 4.7k Back	Pull-up keeping FLASH_DQ3 high when it is not driven. Supports inactive flash-control state. Order: TBD_0402_1pct_4.7k	R102.1: VCCAUX_1V8 [P] R102.2: FLASH_DQ3 [C]
R103 33 Back	Series damping candidate between the FPGA and flash nets. Value and timing still require validation. Order: TBD_0402_1pct_33	R103.1: FPGA_CCLK [C] R103.2: FLASH_SCLK [C]
R104 0 Back	Zero-ohm series link / tuning position between the FPGA and flash nets. Value and timing still require validation. Order: TBD_0402_1pct_0	R104.1: FPGA_CFG_DQ0 [C] R104.2: FLASH_DQ0 [C]
R105 33 Back	Series damping candidate between the FPGA and flash nets. Value and timing still require validation. Order: TBD_0402_1pct_33	R105.1: FPGA_CFG_DQ1 [C] R105.2: FLASH_DQ1 [C]
R106 0 Back	Zero-ohm series link / tuning position between the FPGA and flash nets. Value and timing still require validation. Order: TBD_0402_1pct_0	R106.1: FPGA_CFG_DQ2 [C] R106.2: FLASH_DQ2 [C]
R107 0 Back	Zero-ohm series link / tuning position between the FPGA and flash nets. Value and timing still require validation. Order: TBD_0402_1pct_0	R107.1: FPGA_CFG_DQ3 [C] R107.2: FLASH_DQ3 [C]
R108 4.7k Front	Pull-up for the FPGA FPGA_PROGRAM_B control/status node. Order: TBD_0402_1pct_4.7k	R108.1: VCCAUX_1V8 [P] R108.2: FPGA_PROGRAM_B [C]
R109 4.7k Front	Pull-up for the FPGA FPGA_INIT_B control/status node. Order: TBD_0402_1pct_4.7k	R109.1: VCCAUX_1V8 [P] R109.2: FPGA_INIT_B [C]

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Ref. / value / face	Why it is present / saved order code	Endpoint: net [status]
R110 4.7k Front	Pull-up for the FPGA FPGA_DONE control/status node. Order: TBD_0402_1pct_4.7k	R110.1: VCCAUX_1V8 [P] R110.2: FPGA_DONE [C]
R111 1k Front	Configuration-mode strap: pull high. The three straps encode M[2:0]=001 in this revision. Order: TBD_0402_1pct_1k	R111.1: VCCAUX_1V8 [P] R111.2: CFG_M0 [C]
R112 1k Front	Configuration-mode strap: pull low. The three straps encode M[2:0]=001 in this revision. Order: TBD_0402_1pct_1k	R112.1: CFG_M1 [C] R112.2: GND [P]
R113 1k Front	Configuration-mode strap: pull low. The three straps encode M[2:0]=001 in this revision. Order: TBD_0402_1pct_1k	R113.1: CFG_M2 [C] R113.2: GND [P]
R114 1k Front	Pulls PUDC_B high to disable configuration-time user-I/O pull-ups. External ASIC safe-state handling remains separate. Order: TBD_0402_1pct_1k	R114.1: VCCAUX_1V8 [P] R114.2: CFG_PUDC_B [C]
R115 10k Front	JTAG input pull-up on JTAG_TMS. Cable drive, voltage and ringing still require qualification. Order: TBD_0402_1pct_10k	R115.1: VCCAUX_1V8 [P] R115.2: JTAG_TMS [C]
R116 10k Front	JTAG input pull-up on JTAG_TDI. Cable drive, voltage and ringing still require qualification. Order: TBD_0402_1pct_10k	R116.1: VCCAUX_1V8 [P] R116.2: JTAG_TDI [C]
R117 10k Front	JTAG input pull-up on JTAG_TCK. Cable drive, voltage and ringing still require qualification. Order: TBD_0402_1pct_10k	R117.1: VCCAUX_1V8 [P] R117.2: JTAG_TCK [C]
R118 33 Back	Series damping candidate on FPGA TDO before J4.18. It is not a qualified long-cable termination. Order: TBD_0402_1pct_33	R118.1: FPGA_TDO [C] R118.2: JTAG_TDO [C]
R119 33 Front	Series damping candidate at oscillator output; connects OSC_32M_RAW to CLK_32MHZ. Order: TBD_0402_1pct_33	R119.1: OSC_32M_RAW [C] R119.2: CLK_32MHZ [C]
R122 15m 1% 1W Back	15 milliohm series isolation between local AUX regulator/Cout and the distributed AUX/configuration capacitor network; do not bypass. Order: WSLP1206R0150FEA	R122.1: VAUX_REG_1V803 [P] R122.2: VCCAUX_1V8 [P]
U1 XC7A100T-CSG324 Front	FPGA logic, memory and I/O. Intended to capture the ASIC outputs and control the downstream link; application assignment and firmware remain open. Order: XC7A100T-speedandtemperaturegradeTBD	All 324 balls: Appendix B.
U2 TPS62135RGXR Back	Core buck converter; senses the local VCORE_REG_1V025 node before R9. Order: TPS62135RGXR	All device pins: Appendix D.
U3 TPS62135RGXR Front	1.8 V AUX/configuration converter. Local VOS/C9 is upstream of R122; downstream supply also feeds flash, clock and bank14. Order: TPS62135RGXR	All device pins: Appendix D.
U4 TPS62135RGXR Front	2.5 V bank16/link converter. VSEL is assigned ground but its physical strap remains open. Order: TPS62135RGXR	All device pins: Appendix D.
U5 TPS62135RGXR Back	FPGA ASIC-bank 1.5 V buck converter; enabled by PG_AUX. This is not proof of power delivery to the external ASIC boards. Order: TPS62135RGXR	All device pins: Appendix D.
U6 MX25U12835FM2I-10G Back	1.8 V serial configuration flash for boot storage; signal nets routed, supply/ground network still incomplete. Order: MX25U12835FM2I-10G	All device pins: Appendix D.
Y1 ASE3 32MHz 1.8V Front	1.8 V, 32 MHz reference oscillator; output net routed, supply/ground network still incomplete. Order: ASE3-32.000MHZ-L-C-T	All device pins: Appendix D.

B All 324 FPGA balls

Functions/banks are the saved AMD package-table fields; current nets/status come from the native readback. No ASIC ball allocation has been invented. C = entire assigned net connected; P = assigned net has opens elsewhere or at this endpoint; U = unassigned; NC = intentional no-connect.

Endpoint	Bank	AMD function	Actual net / state	St.
U1.A1	35	IO_L9N_T1_DQS_AD7N_35	Unassigned	U
U1.A2	NA	GND	GND	P
U1.A3	35	IO_L8N_T1_AD14N_35	Unassigned	U
U1.A4	35	IO_L8P_T1_AD14P_35	Unassigned	U
U1.A5	35	IO_L3N_T0_DQS_AD5N_35	Unassigned	U
U1.A6	35	IO_L3P_T0_DQS_AD5P_35	Unassigned	U
U1.A7	35	VCCO_35	VCC_ASIC_1V5	P
U1.A8	16	IO_L12N_T1_MRCC_16	Unassigned	U
U1.A9	16	IO_L14N_T2_SRCC_16	Unassigned	U
U1.A10	16	IO_L14P_T2_SRCC_16	Unassigned	U
U1.A11	15	IO_L4N_T0_15	Unassigned	U
U1.A12	NA	GND	GND	P
U1.A13	15	IO_L9P_T1_DQS_AD3P_15	Unassigned	U
U1.A14	15	IO_L9N_T1_DQS_AD3N_15	Unassigned	U
U1.A15	15	IO_L8P_T1_AD10P_15	Unassigned	U
U1.A16	15	IO_L8N_T1_AD10N_15	Unassigned	U
U1.A17	15	VCCO_15	VCC_ASIC_1V5	P
U1.A18	15	IO_L10N_T1_AD11N_15	Unassigned	U
U1.B1	35	IO_L9P_T1_DQS_AD7P_35	Unassigned	U
U1.B2	35	IO_L10N_T1_AD15N_35	Unassigned	U
U1.B3	35	IO_L10P_T1_AD15P_35	Unassigned	U
U1.B4	35	IO_L7N_T1_AD6N_35	Unassigned	U
U1.B5	NA	GND	GND	P
U1.B6	35	IO_L2N_T0_AD12N_35	Unassigned	U
U1.B7	35	IO_L2P_T0_AD12P_35	Unassigned	U
U1.B8	16	IO_L12P_T1_MRCC_16	Unassigned	U
U1.B9	16	IO_L11N_T1_SRCC_16	Unassigned	U
U1.B10	16	VCCO_16	VCC_LINK_2V5	P
U1.B11	15	IO_L4P_T0_15	Unassigned	U
U1.B12	15	IO_L3N_T0_DQS_AD1N_15	Unassigned	U
U1.B13	15	IO_L2P_T0_AD8P_15	Unassigned	U
U1.B14	15	IO_L2N_T0_AD8N_15	Unassigned	U
U1.B15	NA	GND	GND	P
U1.B16	15	IO_L7P_T1_AD2P_15	Unassigned	U
U1.B17	15	IO_L7N_T1_AD2N_15	Unassigned	U
U1.B18	15	IO_L10P_T1_AD11P_15	Unassigned	U
U1.C1	35	IO_L16N_T2_35	Unassigned	U
U1.C2	35	IO_L16P_T2_35	Unassigned	U
U1.C3	35	VCCO_35	VCC_ASIC_1V5	P
U1.C4	35	IO_L7P_T1_AD6P_35	Unassigned	U
U1.C5	35	IO_L1N_T0_AD4N_35	Unassigned	U
U1.C6	35	IO_L1P_T0_AD4P_35	Unassigned	U
U1.C7	35	IO_L4N_T0_35	Unassigned	U
U1.C8	NA	GND	GND	P
U1.C9	16	IO_L11P_T1_SRCC_16	Unassigned	U
U1.C10	16	IO_L13N_T2_MRCC_16	Unassigned	U
U1.C11	16	IO_L13P_T2_MRCC_16	Unassigned	U
U1.C12	15	IO_L3P_T0_DQS_AD1P_15	Unassigned	U
U1.C13	15	VCCO_15	VCC_ASIC_1V5	P
U1.C14	15	IO_L1N_T0_AD0N_15	Unassigned	U
U1.C15	15	IO_L12N_T1_MRCC_15	Unassigned	U

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Endpoint	Bank	AMD function	Actual net / state	St.
U1.C16	15	IO_L20P_T3_A20_15	Unassigned	U
U1.C17	15	IO_L20N_T3_A19_15	Unassigned	U
U1.C18	NA	GND	GND	P
U1.D1	NA	GND	GND	P
U1.D2	35	IO_L14N_T2_SRCC_35	Unassigned	U
U1.D3	35	IO_L12N_T1_MRCC_35	Unassigned	U
U1.D4	35	IO_L11N_T1_SRCC_35	Unassigned	U
U1.D5	35	IO_L11P_T1_SRCC_35	Unassigned	U
U1.D6	35	VCCO_35	VCC_ASIC_1V5	P
U1.D7	35	IO_L6N_T0_VREF_35	Unassigned	U
U1.D8	35	IO_L4P_T0_35	Unassigned	U
U1.D9	16	IO_L6N_T0_VREF_16	Unassigned	U
U1.D10	16	IO_L19N_T3_VREF_16	Unassigned	U
U1.D11	NA	GND	GND	P
U1.D12	15	IO_L6P_T0_15	Unassigned	U
U1.D13	15	IO_L6N_T0_VREF_15	Unassigned	U
U1.D14	15	IO_L1P_T0_AD0P_15	Unassigned	U
U1.D15	15	IO_L12P_T1_MRCC_15	Unassigned	U
U1.D16	15	VCCO_15	VCC_ASIC_1V5	P
U1.D17	15	IO_L16N_T2_A27_15	Unassigned	U
U1.D18	15	IO_L21N_T3_DQS_A18_15	Unassigned	U
U1.E1	35	IO_L18N_T2_35	Unassigned	U
U1.E2	35	IO_L14P_T2_SRCC_35	Unassigned	U
U1.E3	35	IO_L12P_T1_MRCC_35	Unassigned	U
U1.E4	NA	GND	GND	P
U1.E5	35	IO_L5N_T0_AD13N_35	Unassigned	U
U1.E6	35	IO_L5P_T0_AD13P_35	Unassigned	U
U1.E7	35	IO_L6P_T0_35	Unassigned	U
U1.E8	0	VCCBATT_0	GND	P
U1.E9	0	CCLK_0	FPGA_CCLK	C
U1.E10	0	TCK_0	JTAG_TCK	C
U1.E11	0	TDI_0	JTAG_TDI	C
U1.E12	0	TMS_0	JTAG_TMS	C
U1.E13	0	TDO_0	FPGA_TDO	C
U1.E14	NA	GND	GND	P
U1.E15	15	IO_L11P_T1_SRCC_15	Unassigned	U
U1.E16	15	IO_L11N_T1_SRCC_15	Unassigned	U
U1.E17	15	IO_L16P_T2_A28_15	Unassigned	U
U1.E18	15	IO_L21P_T3_DQS_15	Unassigned	U
U1.F1	35	IO_L18P_T2_35	Unassigned	U
U1.F2	35	VCCO_35	VCC_ASIC_1V5	P
U1.F3	35	IO_L13N_T2_MRCC_35	Unassigned	U
U1.F4	35	IO_L13P_T2_MRCC_35	Unassigned	U
U1.F5	35	IO_0_35	Unassigned	U
U1.F6	35	IO_L19N_T3_VREF_35	Unassigned	U
U1.F7	NA	GND	GND	P
U1.F8	NA	VCCINT	VCCINT_1V0	P
U1.F9	NA	GND	GND	P
U1.F10	NA	VCCBRAM	VCCINT_1V0	P
U1.F11	NA	GND	GND	P
U1.F12	NA	VCCAUX	VCCAUX_1V8	P
U1.F13	15	IO_L5P_T0_AD9P_15	Unassigned	U
U1.F14	15	IO_L5N_T0_AD9N_15	Unassigned	U
U1.F15	15	IO_L14P_T2_SRCC_15	Unassigned	U
U1.F16	15	IO_L14N_T2_SRCC_15	Unassigned	U
U1.F17	NA	GND	GND	P
U1.F18	15	IO_L22N_T3_A16_15	Unassigned	U
U1.G1	35	IO_L17N_T2_35	Unassigned	U

Continued on next page

Endpoint	Bank	AMD function	Actual net / state	St.
U1.G2	35	IO_L15N_T2_DQS_35	Unassigned	U
U1.G3	35	IO_L20N_T3_35	Unassigned	U
U1.G4	35	IO_L20P_T3_35	Unassigned	U
U1.G5	35	VCCO_35	VCC_ASIC_1V5	P
U1.G6	35	IO_L19P_T3_35	Unassigned	U
U1.G7	NA	VCCINT	VCCINT_1V0	P
U1.G8	NA	GND	GND	P
U1.G9	NA	VCCINT	VCCINT_1V0	P
U1.G10	NA	GND	GND	P
U1.G11	NA	VCCBRAM	VCCINT_1V0	P
U1.G12	NA	GND	GND	P
U1.G13	15	IO_0_15	Unassigned	U
U1.G14	15	IO_L15N_T2_DQS_ADV_B_15	Unassigned	U
U1.G15	15	VCCO_15	VCC_ASIC_1V5	P
U1.G16	15	IO_L13N_T2_MRCC_15	Unassigned	U
U1.G17	15	IO_L18N_T2_A23_15	Unassigned	U
U1.G18	15	IO_L22P_T3_A17_15	Unassigned	U
U1.H1	35	IO_L17P_T2_35	Unassigned	U
U1.H2	35	IO_L15P_T2_DQS_35	Unassigned	U
U1.H3	NA	GND	GND	P
U1.H4	35	IO_L21N_T3_DQS_35	Unassigned	U
U1.H5	35	IO_L24N_T3_35	Unassigned	U
U1.H6	35	IO_L24P_T3_35	Unassigned	U
U1.H7	NA	GND	GND	P
U1.H8	NA	VCCINT	VCCINT_1V0	P
U1.H9	0	GNDADC_0	GND	P
U1.H10	0	VCCADC_0	VCCAUX_1V8	P
U1.H11	NA	GND	GND	P
U1.H12	NA	VCCAUX	VCCAUX_1V8	P
U1.H13	NA	GND	GND	P
U1.H14	15	IO_L15P_T2_DQS_15	Unassigned	U
U1.H15	15	IO_L19N_T3_A21_VREF_15	Unassigned	U
U1.H16	15	IO_L13P_T2_MRCC_15	Unassigned	U
U1.H17	15	IO_L18P_T2_A24_15	Unassigned	U
U1.H18	15	VCCO_15	VCC_ASIC_1V5	P
U1.J1	35	VCCO_35	VCC_ASIC_1V5	P
U1.J2	35	IO_L22N_T3_35	Unassigned	U
U1.J3	35	IO_L22P_T3_35	Unassigned	U
U1.J4	35	IO_L21P_T3_DQS_35	Unassigned	U
U1.J5	35	IO_25_35	Unassigned	U
U1.J6	NA	GND	GND	P
U1.J7	NA	VCCINT	VCCINT_1V0	P
U1.J8	NA	GND	GND	P
U1.J9	0	VREFN_0	GND	P
U1.J10	0	VP_0	GND	P
U1.J11	NA	VCCINT	VCCINT_1V0	P
U1.J12	NA	GND	GND	P
U1.J13	15	IO_L17N_T2_A25_15	Unassigned	U
U1.J14	15	IO_L19P_T3_A22_15	Unassigned	U
U1.J15	15	IO_L24N_T3_RS0_15	Unassigned	U
U1.J16	NA	GND	GND	P
U1.J17	15	IO_L23P_T3_FOE_B_15	Unassigned	U
U1.J18	15	IO_L23N_T3_FWE_B_15	Unassigned	U
U1.K1	35	IO_L23N_T3_35	Unassigned	U
U1.K2	35	IO_L23P_T3_35	Unassigned	U
U1.K3	34	IO_L2P_T0_34	Unassigned	U
U1.K4	34	VCCO_34	VCC_ASIC_1V5	P
U1.K5	34	IO_L5P_T0_34	Unassigned	U

Continued on next page

Endpoint	Bank	AMD function	Actual net / state	St.
U1.K6	34	IO_0_34	Unassigned	U
U1.K7	NA	GND	GND	P
U1.K8	NA	VCCINT	VCCINT_1V0	P
U1.K9	0	VN_0	GND	P
U1.K10	0	VREFP_0	GND	P
U1.K11	NA	GND	GND	P
U1.K12	NA	VCCAUX	VCCAUX_1V8	P
U1.K13	15	IO_L17P_T2_A26_15	Unassigned	U
U1.K14	15	VCCO_15	VCC_ASIC_1V5	P
U1.K15	15	IO_L24P_T3_RS1_15	Unassigned	U
U1.K16	15	IO_25_15	Unassigned	U
U1.K17	14	IO_L1P_T0_D00_MOSI_14	FPGA_CFG_DQ0	C
U1.K18	14	IO_L1N_T0_D01_DIN_14	FPGA_CFG_DQ1	C
U1.L1	34	IO_L1P_T0_34	Unassigned	U
U1.L2	NA	GND	GND	P
U1.L3	34	IO_L2N_T0_34	Unassigned	U
U1.L4	34	IO_L5N_T0_34	Unassigned	U
U1.L5	34	IO_L6N_T0_VREF_34	Unassigned	U
U1.L6	34	IO_L6P_T0_34	Unassigned	U
U1.L7	NA	VCCINT	VCCINT_1V0	P
U1.L8	NA	GND	GND	P
U1.L9	0	DXN_0	Intentional NC	NC
U1.L10	0	DXP_0	Intentional NC	NC
U1.L11	NA	VCCINT	VCCINT_1V0	P
U1.L12	NA	GND	GND	P
U1.L13	14	IO_L6P_T0_FCS_B_14	FLASH_CS_B	C
U1.L14	14	IO_L2P_T0_D02_14	FPGA_CFG_DQ2	C
U1.L15	14	IO_L3P_T0_DQS_PUDC_B_14	CFG_PUDC_B	C
U1.L16	14	IO_L3N_T0_DQS_EMCCLK_14	Unassigned	U
U1.L17	14	VCCO_14	VCCAUX_1V8	P
U1.L18	14	IO_L4P_T0_D04_14	Unassigned	U
U1.M1	34	IO_L1N_T0_34	Unassigned	U
U1.M2	34	IO_L4N_T0_34	Unassigned	U
U1.M3	34	IO_L4P_T0_34	Unassigned	U
U1.M4	34	IO_L16P_T2_34	Unassigned	U
U1.M5	NA	GND	GND	P
U1.M6	34	IO_L18P_T2_34	Unassigned	U
U1.M7	NA	GND	GND	P
U1.M8	NA	VCCINT	VCCINT_1V0	P
U1.M9	NA	GND	GND	P
U1.M10	NA	VCCINT	VCCINT_1V0	P
U1.M11	NA	GND	GND	P
U1.M12	NA	VCCAUX	VCCAUX_1V8	P
U1.M13	14	IO_L6N_T0_D08_VREF_14	Unassigned	U
U1.M14	14	IO_L2N_T0_D03_14	FPGA_CFG_DQ3	C
U1.M15	NA	GND	GND	P
U1.M16	14	IO_L10P_T1_D14_14	Unassigned	U
U1.M17	14	IO_L10N_T1_D15_14	Unassigned	U
U1.M18	14	IO_L4N_T0_D05_14	Unassigned	U
U1.N1	34	IO_L3N_T0_DQS_34	Unassigned	U
U1.N2	34	IO_L3P_T0_DQS_34	Unassigned	U
U1.N3	34	VCCO_34	VCC_ASIC_1V5	P
U1.N4	34	IO_L16N_T2_34	Unassigned	U
U1.N5	34	IO_L13P_T2_MRCC_34	Unassigned	U
U1.N6	34	IO_L18N_T2_34	Unassigned	U
U1.N7	NA	VCCINT	VCCINT_1V0	P
U1.N8	NA	GND	GND	P
U1.N9	NA	VCCINT	VCCINT_1V0	P

Continued on next page

Endpoint	Bank	AMD function	Actual net / state	St.
U1.N10	NA	GND	GND	P
U1.N11	NA	VCCINT	VCCINT_1V0	P
U1.N12	NA	GND	GND	P
U1.N13	14	VCCO_14	VCCAUX_1V8	P
U1.N14	14	IO_L8P_T1_D11_14	Unassigned	U
U1.N15	14	IO_L11P_T1_SRCC_14	Unassigned	U
U1.N16	14	IO_L11N_T1_SRCC_14	Unassigned	U
U1.N17	14	IO_L9P_T1_DQS_14	Unassigned	U
U1.N18	NA	GND	GND	P
U1.P1	NA	GND	GND	P
U1.P2	34	IO_L15P_T2_DQS_34	Unassigned	U
U1.P3	34	IO_L14N_T2_SRCC_34	Unassigned	U
U1.P4	34	IO_L14P_T2_SRCC_34	Unassigned	U
U1.P5	34	IO_L13N_T2_MRCC_34	Unassigned	U
U1.P6	34	VCCO_34	VCC_ASIC_1V5	P
U1.P7	0	INIT_B_0	FPGA_INIT_B	C
U1.P8	0	CFGBVS_0	GND	P
U1.P9	0	PROGRAM_B_0	FPGA_PROGRAM_B	C
U1.P10	0	DONE_0	FPGA_DONE	C
U1.P11	0	M2_0	CFG_M2	C
U1.P12	0	M0_0	CFG_M0	C
U1.P13	0	M1_0	CFG_M1	C
U1.P14	14	IO_L8N_T1_D12_14	Unassigned	U
U1.P15	14	IO_L13P_T2_MRCC_14	Unassigned	U
U1.P16	14	VCCO_14	VCCAUX_1V8	P
U1.P17	14	IO_L12P_T1_MRCC_14	CLK_32MHZ	C
U1.P18	14	IO_L9N_T1_DQS_D13_14	Unassigned	U
U1.R1	34	IO_L17P_T2_34	Unassigned	U
U1.R2	34	IO_L15N_T2_DQS_34	Unassigned	U
U1.R3	34	IO_L11P_T1_SRCC_34	Unassigned	U
U1.R4	NA	GND	GND	P
U1.R5	34	IO_L19N_T3_VREF_34	Unassigned	U
U1.R6	34	IO_L19P_T3_34	Unassigned	U
U1.R7	34	IO_L23P_T3_34	Unassigned	U
U1.R8	34	IO_L24P_T3_34	Unassigned	U
U1.R9	0	VCCO_0	VCCAUX_1V8	P
U1.R10	14	IO_25_14	Unassigned	U
U1.R11	14	IO_0_14	Unassigned	U
U1.R12	14	IO_L5P_T0_D06_14	Unassigned	U
U1.R13	14	IO_L5N_T0_D07_14	Unassigned	U
U1.R14	NA	GND	GND	P
U1.R15	14	IO_L13N_T2_MRCC_14	Unassigned	U
U1.R16	14	IO_L15P_T2_DQS_RDWR_B_14	Unassigned	U
U1.R17	14	IO_L12N_T1_MRCC_14	Unassigned	U
U1.R18	14	IO_L7P_T1_D09_14	Unassigned	U
U1.T1	34	IO_L17N_T2_34	Unassigned	U
U1.T2	34	VCCO_34	VCC_ASIC_1V5	P
U1.T3	34	IO_L11N_T1_SRCC_34	Unassigned	U
U1.T4	34	IO_L12N_T1_MRCC_34	Unassigned	U
U1.T5	34	IO_L12P_T1_MRCC_34	Unassigned	U
U1.T6	34	IO_L23N_T3_34	Unassigned	U
U1.T7	NA	GND	GND	P
U1.T8	34	IO_L24N_T3_34	Unassigned	U
U1.T9	14	IO_L24P_T3_A01_D17_14	Unassigned	U
U1.T10	14	IO_L24N_T3_A00_D16_14	Unassigned	U
U1.T11	14	IO_L19P_T3_A10_D26_14	Unassigned	U
U1.T12	14	VCCO_14	VCCAUX_1V8	P
U1.T13	14	IO_L23P_T3_A03_D19_14	Unassigned	U

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Endpoint	Bank	AMD function	Actual net / state	St.
U1.T14	14	IO_L14P_T2_SRCC_14	Unassigned	U
U1.T15	14	IO_L14N_T2_SRCC_14	Unassigned	U
U1.T16	14	IO_L15N_T2_DQS_DOUT_CSO_B_14	Unassigned	U
U1.T17	NA	GND	GND	P
U1.T18	14	IO_L7N_T1_D10_14	Unassigned	U
U1.U1	34	IO_L7P_T1_34	Unassigned	U
U1.U2	34	IO_L9P_T1_DQS_34	Unassigned	U
U1.U3	34	IO_L8N_T1_34	Unassigned	U
U1.U4	34	IO_L8P_T1_34	Unassigned	U
U1.U5	34	VCCO_34	VCC_ASIC_1V5	P
U1.U6	34	IO_L22N_T3_34	Unassigned	U
U1.U7	34	IO_L22P_T3_34	Unassigned	U
U1.U8	34	IO_25_34	Unassigned	U
U1.U9	34	IO_L21P_T3_DQS_34	Unassigned	U
U1.U10	NA	GND	GND	P
U1.U11	14	IO_L19N_T3_A09_D25_VREF_14	Unassigned	U
U1.U12	14	IO_L20P_T3_A08_D24_14	Unassigned	U
U1.U13	14	IO_L23N_T3_A02_D18_14	Unassigned	U
U1.U14	14	IO_L22P_T3_A05_D21_14	Unassigned	U
U1.U15	14	VCCO_14	VCCAUX_1V8	P
U1.U16	14	IO_L18P_T2_A12_D28_14	Unassigned	U
U1.U17	14	IO_L17P_T2_A14_D30_14	Unassigned	U
U1.U18	14	IO_L17N_T2_A13_D29_14	Unassigned	U
U1.V1	34	IO_L7N_T1_34	Unassigned	U
U1.V2	34	IO_L9N_T1_DQS_34	Unassigned	U
U1.V3	NA	GND	GND	P
U1.V4	34	IO_L10N_T1_34	Unassigned	U
U1.V5	34	IO_L10P_T1_34	Unassigned	U
U1.V6	34	IO_L20N_T3_34	Unassigned	U
U1.V7	34	IO_L20P_T3_34	Unassigned	U
U1.V8	34	VCCO_34	VCC_ASIC_1V5	P
U1.V9	34	IO_L21N_T3_DQS_34	Unassigned	U
U1.V10	14	IO_L21P_T3_DQS_14	Unassigned	U
U1.V11	14	IO_L21N_T3_DQS_A06_D22_14	Unassigned	U
U1.V12	14	IO_L20N_T3_A07_D23_14	Unassigned	U
U1.V13	NA	GND	GND	P
U1.V14	14	IO_L22N_T3_A04_D20_14	Unassigned	U
U1.V15	14	IO_L16P_T2_CSI_B_14	Unassigned	U
U1.V16	14	IO_L16N_T2_A15_D31_14	Unassigned	U
U1.V17	14	IO_L18N_T2_A11_D27_14	Unassigned	U
U1.V18	14	VCCO_14	VCCAUX_1V8	P

C External connector pins

C.1 J4: custom cable interface

This micro-HDMI-style connector carries a custom power/JTAG/reserved-data assignment. Pin 19 is 12 V, pin 1 is the 1.8 V target reference, and pin 16 is a dedicated ground return. It is incompatible with ordinary HDMI equipment. The eight reserved signal contacts are unassigned. Four physical shell pads share SH.

Endpoint	Role / function	Actual net / state	St.	Pads
J4.1	VTREF	VCCAUX_1V8	P	1
J4.2	JTAG_TMS	JTAG_TMS	C	1
J4.3	PAIR0_P_RESERVED	Unassigned	U	1
J4.4	GND	GND	P	1
J4.5	PAIR0_N_RESERVED	Unassigned	U	1
J4.6	PAIR1_P_RESERVED	Unassigned	U	1
J4.7	GND	GND	P	1
J4.8	PAIR1_N_RESERVED	Unassigned	U	1
J4.9	PAIR2_P_RESERVED	Unassigned	U	1
J4.10	GND	GND	P	1
J4.11	PAIR2_N_RESERVED	Unassigned	U	1
J4.12	PAIR3_P_RESERVED	Unassigned	U	1
J4.13	GND	GND	P	1
J4.14	PAIR3_N_RESERVED	Unassigned	U	1
J4.15	JTAG_TDI	JTAG_TDI	C	1
J4.16	GND	GND	P	1
J4.17	JTAG_TCK	JTAG_TCK	C	1
J4.18	JTAG_TDO	JTAG_TDO	C	1
J4.19	12V	LINK_12V	P	1
J4.SH	GND	GND	P	4

C.2 J5 and J6: actual contacts and the separate proposal

All 120 signal contacts are still unassigned in the PCB. Proposed names below are planning references, not routed connections or safe voltage certifications. Each connector's four physical G lands share GND; two unnumbered locating holes are not electrical pins. The GND net is only partially completed.

Endpoint	Actual PCB state	Unimplemented proposed signal	Proposed direction
J5.1	Unassigned [U]	ASIC1_DATA1	Input
J5.2	Unassigned [U]	ASIC1_DATA2	Input
J5.3	Unassigned [U]	ASIC1_DATA3	Input
J5.4	Unassigned [U]	ASIC1_DATA4	Input
J5.5	Unassigned [U]	ASIC1_DATA5	Input
J5.6	Unassigned [U]	ASIC1_DATA6	Input
J5.7	Unassigned [U]	ASIC1_DATA7	Input
J5.8	Unassigned [U]	ASIC1_DATA8	Input
J5.9	Unassigned [U]	ASIC1_CLK32MHz_Out	Input
J5.10	Unassigned [U]	ASIC1_READ	Input
J5.11	Unassigned [U]	ASIC1_SYNC	Input
J5.12	Unassigned [U]	ASIC2_DATA1	Input
J5.13	Unassigned [U]	ASIC2_DATA2	Input
J5.14	Unassigned [U]	ASIC2_DATA3	Input
J5.15	Unassigned [U]	ASIC2_DATA4	Input
J5.16	Unassigned [U]	ASIC2_DATA5	Input

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Endpoint	Actual PCB state	Unimplemented proposed signal	Proposed direction
J5.17	Unassigned [U]	ASIC2_DATA6	Input
J5.18	Unassigned [U]	ASIC2_DATA7	Input
J5.19	Unassigned [U]	ASIC2_DATA8	Input
J5.20	Unassigned [U]	ASIC2_CLK32MHz_Out	Input
J5.21	Unassigned [U]	ASIC2_READ	Input
J5.22	Unassigned [U]	ASIC2_SYNC	Input
J5.23	Unassigned [U]	ASIC3_DATA1	Input
J5.24	Unassigned [U]	ASIC3_DATA2	Input
J5.25	Unassigned [U]	ASIC3_DATA3	Input
J5.26	Unassigned [U]	ASIC3_DATA4	Input
J5.27	Unassigned [U]	ASIC3_DATA5	Input
J5.28	Unassigned [U]	ASIC3_DATA6	Input
J5.29	Unassigned [U]	ASIC3_DATA7	Input
J5.30	Unassigned [U]	ASIC3_DATA8	Input
J5.31	Unassigned [U]	ASIC3_CLK32MHz_Out	Input
J5.32	Unassigned [U]	ASIC3_READ	Input
J5.33	Unassigned [U]	ASIC3_SYNC	Input
J5.34	Unassigned [U]	ASIC4_DATA1	Input
J5.35	Unassigned [U]	ASIC4_DATA2	Input
J5.36	Unassigned [U]	ASIC4_DATA3	Input
J5.37	Unassigned [U]	ASIC4_DATA4	Input
J5.38	Unassigned [U]	ASIC4_DATA5	Input
J5.39	Unassigned [U]	ASIC4_DATA6	Input
J5.40	Unassigned [U]	ASIC4_DATA7	Input
J5.41	Unassigned [U]	ASIC4_DATA8	Input
J5.42	Unassigned [U]	ASIC4_CLK32MHz_Out	Input
J5.43	Unassigned [U]	ASIC4_READ	Input
J5.44	Unassigned [U]	ASIC4_SYNC	Input
J5.45	Unassigned [U]	CHIP_RESET_SHARED	Output
J5.46	Unassigned [U]	AC_IN_SHARED	TBD: verify circuit role
J5.47	Unassigned [U]	IMP_TST_SHARED	TBD: verify circuit role
J5.48	Unassigned [U]	FE_RESET_SHARED	Output
J5.49	Unassigned [U]	SPI_LATCH_SHARED	Output
J5.50	Unassigned [U]	STIM_CLK_SHARED	Output
J5.51	Unassigned [U]	STIM_START_SHARED	Output
J5.52	Unassigned [U]	STIM_EN_SHARED	Output
J5.53	Unassigned [U]	STIM_CHB_SHARED	Output
J5.54	Unassigned [U]	BOARD1_CLK	Output
J5.55	Unassigned [U]	BOARD2_CLK	Output
J5.56	Unassigned [U]	BOARD3_CLK	Output
J5.57	Unassigned [U]	BOARD4_CLK	Output
J5.58	Unassigned [U]	BOARD1_SPI_CLK	Output
J5.59	Unassigned [U]	BOARD2_SPI_CLK	Output
J5.60	Unassigned [U]	BOARD3_SPI_CLK	Output
J5.G	GND [P]	GND	Ground reference
J6.1	Unassigned [U]	ASIC5_DATA1	Input
J6.2	Unassigned [U]	ASIC5_DATA2	Input
J6.3	Unassigned [U]	ASIC5_DATA3	Input
J6.4	Unassigned [U]	ASIC5_DATA4	Input
J6.5	Unassigned [U]	ASIC5_DATA5	Input
J6.6	Unassigned [U]	ASIC5_DATA6	Input
J6.7	Unassigned [U]	ASIC5_DATA7	Input
J6.8	Unassigned [U]	ASIC5_DATA8	Input
J6.9	Unassigned [U]	ASIC5_CLK32MHz_Out	Input
J6.10	Unassigned [U]	ASIC5_READ	Input
J6.11	Unassigned [U]	ASIC5_SYNC	Input
J6.12	Unassigned [U]	ASIC6_DATA1	Input
J6.13	Unassigned [U]	ASIC6_DATA2	Input

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Endpoint	Actual PCB state	Unimplemented proposed signal	Proposed direction
J6.14	Unassigned [U]	ASIC6_DATA3	Input
J6.15	Unassigned [U]	ASIC6_DATA4	Input
J6.16	Unassigned [U]	ASIC6_DATA5	Input
J6.17	Unassigned [U]	ASIC6_DATA6	Input
J6.18	Unassigned [U]	ASIC6_DATA7	Input
J6.19	Unassigned [U]	ASIC6_DATA8	Input
J6.20	Unassigned [U]	ASIC6_CLK32MHz_Out	Input
J6.21	Unassigned [U]	ASIC6_READ	Input
J6.22	Unassigned [U]	ASIC6_SYNC	Input
J6.23	Unassigned [U]	ASIC7_DATA1	Input
J6.24	Unassigned [U]	ASIC7_DATA2	Input
J6.25	Unassigned [U]	ASIC7_DATA3	Input
J6.26	Unassigned [U]	ASIC7_DATA4	Input
J6.27	Unassigned [U]	ASIC7_DATA5	Input
J6.28	Unassigned [U]	ASIC7_DATA6	Input
J6.29	Unassigned [U]	ASIC7_DATA7	Input
J6.30	Unassigned [U]	ASIC7_DATA8	Input
J6.31	Unassigned [U]	ASIC7_CLK32MHz_Out	Input
J6.32	Unassigned [U]	ASIC7_READ	Input
J6.33	Unassigned [U]	ASIC7_SYNC	Input
J6.34	Unassigned [U]	ASIC8_DATA1	Input
J6.35	Unassigned [U]	ASIC8_DATA2	Input
J6.36	Unassigned [U]	ASIC8_DATA3	Input
J6.37	Unassigned [U]	ASIC8_DATA4	Input
J6.38	Unassigned [U]	ASIC8_DATA5	Input
J6.39	Unassigned [U]	ASIC8_DATA6	Input
J6.40	Unassigned [U]	ASIC8_DATA7	Input
J6.41	Unassigned [U]	ASIC8_DATA8	Input
J6.42	Unassigned [U]	ASIC8_CLK32MHz_Out	Input
J6.43	Unassigned [U]	ASIC8_READ	Input
J6.44	Unassigned [U]	ASIC8_SYNC	Input
J6.45	Unassigned [U]	BOARD4_SPI_CLK	Output
J6.46	Unassigned [U]	STIM_CHIP1_SPI_DL	Output
J6.47	Unassigned [U]	STIM_CHIP1_SPI_DR	Output
J6.48	Unassigned [U]	STIM_CHIP2_SPI_DL	Output
J6.49	Unassigned [U]	STIM_CHIP2_SPI_DR	Output
J6.50	Unassigned [U]	STIM_CHIP3_SPI_DL	Output
J6.51	Unassigned [U]	STIM_CHIP3_SPI_DR	Output
J6.52	Unassigned [U]	STIM_CHIP4_SPI_DL	Output
J6.53	Unassigned [U]	STIM_CHIP4_SPI_DR	Output
J6.54	Unassigned [U]	NONSTIM_BOARD1_SPI_DL	Output
J6.55	Unassigned [U]	NONSTIM_BOARD1_SPI_DR	Output
J6.56	Unassigned [U]	NONSTIM_BOARD2_SPI_DL	Output
J6.57	Unassigned [U]	NONSTIM_BOARD2_SPI_DR	Output
J6.58	Unassigned [U]	RESERVED_UNCONNECTED	Reserved
J6.59	Unassigned [U]	RESERVED_UNCONNECTED	Reserved
J6.60	Unassigned [U]	RESERVED_UNCONNECTED	Reserved
J6.G	GND [P]	GND	Ground reference

D Converter, flash and oscillator pins

Every non-FPGA IC/oscillator pin appears below. Pin functions follow the selected-device interpretation, not an assumed generic-library name. U4.11's GND assignment is physically open; all FB2 pins are intentional NC. P means the whole named net remains incomplete, not that this specific pad is necessarily isolated.

Endpoint	Selected-part pin function	Actual net / state	St.
U2.1	VIN	LINK_12V	P
U2.2	SW	SW_CORE	C
U2.3	GND	GND	P
U2.4	FB2 (unused)	Intentional NC	NC
U2.5	FB	FB_CORE	C
U2.6	VOS	VCORE_REG_1V025	C
U2.7	PG	PG_CORE	C
U2.8	EN	LINK_12V	P
U2.9	SS/TR	SS_CORE	C
U2.10	MODE	LINK_12V	P
U2.11	VSEL	GND	P
U3.1	VIN	LINK_12V	P
U3.2	SW	SW_AUX	C
U3.3	GND	GND	P
U3.4	FB2 (unused)	Intentional NC	NC
U3.5	FB	FB_AUX	C
U3.6	VOS	VAUX_REG_1V803	P
U3.7	PG	PG_AUX	C
U3.8	EN	PG_CORE	C
U3.9	SS/TR	SS_AUX	C
U3.10	MODE	LINK_12V	P
U3.11	VSEL	GND	P
U4.1	VIN	LINK_12V	P
U4.2	SW	SW_CFG	C
U4.3	GND	GND	P
U4.4	FB2 (unused)	Intentional NC	NC
U4.5	FB	FB_CFG	C
U4.6	VOS	VCC_LINK_2V5	P
U4.7	PG	PGOOD_IO	C
U4.8	EN	PG_AUX	C
U4.9	SS/TR	SS_CFG	C
U4.10	MODE	LINK_12V	P
U4.11	VSEL	GND	P
U5.1	VIN	LINK_12V	P
U5.2	SW	SW_ASIC	C
U5.3	GND	GND	P
U5.4	FB2 (unused)	Intentional NC	NC
U5.5	FB	FB_ASIC	C
U5.6	VOS	VCC_ASIC_1V5	P
U5.7	PG	PGOOD_IO	C
U5.8	EN	PG_AUX	C
U5.9	SS/TR	SS_ASIC	C
U5.10	MODE	LINK_12V	P
U5.11	VSEL	GND	P
U6.1	CS#	FLASH_CS_B	C
U6.2	SO / SIO1	FLASH_DQ1	C
U6.3	WP# / SIO2	FLASH_DQ2	C
U6.4	GND	GND	P
U6.5	SI / SIO0	FLASH_DQ0	C
U6.6	SCLK	FLASH_SCLK	C
U6.7	RESET# / SIO3	FLASH_DQ3	C
U6.8	VCC	VCCAUX_1V8	P
Y1.1	Enable	VCCAUX_1V8	P
Y1.2	GND	GND	P
Y1.3	Clock output	OSC_32M_RAW	C
Y1.4	VDD	VCCAUX_1V8	P

E Every functional net and all peer endpoints

Endpoints in one row share a logical net. The last column gives remaining native unconnected items on that net. A zero establishes native electrical connectivity, not signal quality, current capacity or operational success. Series parts intentionally separate net names. Duplicate physical ground/shell lands share a single endpoint identifier.

Net	Every peer endpoint	Opens
CFG_M0	R111.2, U1.P12	0
CFG_M1	R112.1, U1.P13	0
CFG_M2	R113.1, U1.P11	0
CFG_PUDC_B	R114.2, U1.L15	0
CLK_32MHZ	R119.2, U1.P17	0
FB_ASIC	R7.2, R8.1, U5.5	0
FB_AUX	R3.2, R4.1, U3.5	0
FB_CFG	R5.2, R6.1, U4.5	0
FB_CORE	R1.2, R2.1, U2.5	0
FLASH_CS_B	R100.2, U1.L13, U6.1	0
FLASH_DQ0	R104.2, U6.5	0
FLASH_DQ1	R105.2, U6.2	0
FLASH_DQ2	R101.2, R106.2, U6.3	0
FLASH_DQ3	R102.2, R107.2, U6.7	0
FLASH_SCLK	R103.2, U6.6	0
FPGA_CCLK	R103.1, U1.E9	0
FPGA_CFG_DQ0	R104.1, U1.K17	0
FPGA_CFG_DQ1	R105.1, U1.K18	0
FPGA_CFG_DQ2	R106.1, U1.L14	0
FPGA_CFG_DQ3	R107.1, U1.M14	0
FPGA_DONE	R110.2, U1.P10	0
FPGA_INIT_B	R109.2, U1.P7	0
FPGA_PROGRAM_B	R108.2, U1.P9	0
FPGA_TDO	R118.1, U1.E13	0
GND	C1.2, C10.2, C100.2, C101.2, C102.2, C103.2, C11.2, C12.2, C13.2, C14.2, C15.2, C16.2, C17.2, C18.2, C2.2, C20.2	8
GND	C21.2, C22.2, C23.2, C24.2, C25.2, C26.2, C27.2, C28.2, C29.2, C3.2, C30.2, C31.2, C32.2, C33.2, C34.2, C35.2	–
GND	C36.2, C37.2, C4.2, C40.2, C41.2, C42.2, C43.2, C44.2, C45.2, C46.2, C5.2, C50.2, C51.2, C52.2, C53.2, C54.2	–
GND	C55.2, C56.2, C57.2, C58.2, C59.2, C6.2, C60.2, C61.2, C62.2, C63.2, C64.2, C65.2, C66.2, C67.2, C68.2, C69.2	–
GND	C7.2, C70.2, C71.2, C72.2, C73.2, C74.2, C75.2, C76.2, C77.2, C78.2, C79.2, C8.2, C80.2, C81.2, C82.2, C9.2	–
GND	C90.2, C91.2, C92.2, J4.10, J4.13, J4.16, J4.4, J4.7, J4.SH, J5.G, J6.G, R112.2, R113.2, R2.2, R4.2, R6.2	–
GND	R8.2, U1.A12, U1.A2, U1.B15, U1.B5, U1.C18, U1.C8, U1.D1, U1.D11, U1.E14, U1.E4, U1.E8, U1.F11, U1.F17, U1.F7, U1.F9	–
GND	U1.G10, U1.G12, U1.G8, U1.H11, U1.H13, U1.H3, U1.H7, U1.H9, U1.J10, U1.J12, U1.J16, U1.J6, U1.J8, U1.J9, U1.K10, U1.K11	–
GND	U1.K7, U1.K9, U1.L12, U1.L2, U1.L8, U1.M11, U1.M15, U1.M5, U1.M7, U1.M9, U1.N10, U1.N12, U1.N18, U1.N8, U1.P1, U1.P8	–
GND	U1.R14, U1.R4, U1.T17, U1.T7, U1.U10, U1.V13, U1.V3, U2.11, U2.3, U3.11, U3.3, U4.11, U4.3, U5.11, U5.3, U6.4	–
GND	Y1.2	–
JTAG_TCK	J4.17, R117.2, U1.E10	0
JTAG_TDI	J4.15, R116.2, U1.E11	0
JTAG_TDO	J4.18, R118.2	0
JTAG_TMS	J4.2, R115.2, U1.E12	0

Continued on next page

Net	Every peer endpoint	Opens
LINK_12V	C1.1, C11.1, C12.1, C15.1, C16.1, C2.1, C3.1, C4.1, C7.1, C8.1, J4.19, R10.1, R11.1, U2.1, U2.10, U2.8	9
LINK_12V (continued)	U3.1, U3.10, U4.1, U4.10, U5.1, U5.10	–
OSC_32M_RAW	R119.1, Y1.3	0
PGOOD_IO	R12.2, U4.7, U5.7	0
PG_AUX	R11.2, U3.7, U4.8, U5.8	0
PG_CORE	R10.2, U2.7, U3.8	0
SS_ASIC	C18.1, U5.9	0
SS_AUX	C10.1, U3.9	0
SS_CFG	C14.1, U4.9	0
SS_CORE	C6.1, U2.9	0
SW_ASIC	L4.1, U5.2	0
SW_AUX	L2.1, U3.2	0
SW_CFG	L3.1, U4.2	0
SW_CORE	L1.1, U2.2	0
VAUX_REG_1V803	C9.1, L2.2, R122.1, R3.1, U3.6	1
VCCAUX_1V8	C100.1, C101.1, C102.1, C103.1, C40.1, C41.1, C42.1, C43.1, C44.1, C45.1, C46.1, C50.1, C51.1, C52.1, C53.1, C54.1	49
VCCAUX_1V8 (continued)	C55.1, C56.1, C90.1, C91.1, C92.1, J4.1, R100.1, R101.1, R102.1, R108.1, R109.1, R110.1, R111.1, R114.1, R115.1, R116.1	–
VCCAUX_1V8 (continued)	R117.1, R12.1, R122.2, U1.F12, U1.H10, U1.H12, U1.K12, U1.L17, U1.M12, U1.N13, U1.P16, U1.R9, U1.T12, U1.U15, U1.V18, U6.8	–
VCCAUX_1V8 (continued)	Y1.1, Y1.4	–
VCCINT_1V0	C20.1, C21.1, C22.1, C23.1, C24.1, C25.1, C26.1, C27.1, C28.1, C29.1, C30.1, C31.1, C32.1, C33.1, C34.1, C35.1	34
VCCINT_1V0 (continued)	C36.1, C37.1, R9.2, U1.F10, U1.F8, U1.G11, U1.G7, U1.G9, U1.H8, U1.J11, U1.J7, U1.K8, U1.L11, U1.L7, U1.M10, U1.M8	–
VCCINT_1V0 (continued)	U1.N11, U1.N7, U1.N9	–
VCC_ASIC_1V5	C17.1, C64.1, C65.1, C66.1, C67.1, C68.1, C69.1, C70.1, C71.1, C72.1, C73.1, C74.1, C75.1, C76.1, C77.1, C78.1	37
VCC_ASIC_1V5 (continued)	C79.1, C80.1, C81.1, C82.1, L4.2, R7.1, U1.A17, U1.A7, U1.C13, U1.C3, U1.D16, U1.D6, U1.F2, U1.G15, U1.G5, U1.H18	–
VCC_ASIC_1V5 (continued)	U1.J1, U1.K14, U1.K4, U1.N3, U1.P6, U1.T2, U1.U5, U1.V8, U5.6	–
VCC_LINK_2V5	C13.1, C57.1, C58.1, C59.1, C60.1, C61.1, C62.1, C63.1, L3.2, R5.1, U1.B10, U4.6	8
VCORE_REG_1V025	C5.1, L1.2, R1.1, R9.1, U2.6	0

F Evidence, reproducibility and sources

F.1 Exact revision and coverage

Native board SHA-256:

ae601df3925e5ec5e90c8a2e747356e432d0558e820199cf7c01c6bf8a7fef96

The source bundle includes this self-contained LaTeX file, current component/pin/net CSVs, routing snapshot, native readback, verification evidence and build instructions. No external image or additional TeX file is required by the source; the vector map is embedded directly. The earlier dated placement report is preserved unchanged.

Appendix A covers 240 endpoints on 120 two-terminal components; Appendix B covers 324 FPGA balls; Appendix C covers 142 distinct connector endpoints; Appendix D covers 56 other device pins. This totals **762 endpoints**, with no omissions or duplicate entries. All 130 component identifiers and all 48 functional net names are included. Appendix E repeats endpoint identifiers only to expand net membership.

The report was generated from frozen readback/CSV evidence. The native checks preserve the saved rules and report no ignored/excluded checks. PCB parity is not a substitute for the unresolved schematic/ERC issues, electrical contracts or prototype measurements. A net marked C has a geometric connection, not a bench validation.

F.2 Primary references

- [1] Texas Instruments, *TPS62135/TPS621351*, SLVSBH3B, pin table p3; PG p9; discharge p12; distributed capacitance pp32–33; layout p36. <https://www.ti.com/lit/ds/symlink/tps62135.pdf>
- [2] AMD, *Artix 7 DC/AC characteristics*, DS181 v1.27.1, sequencing p8 and ramp limits p9. <https://docs.amd.com/api/khub/documents/iAkxxTOK96ANLJqYf2hgrQ/content>
- [3] AMD, *7 Series Configuration User Guide*, UG470, configuration-bank voltage/CFGBVS and SPI boot guidance. https://docs.amd.com/v/u/en-US/ug470_7Series_Config
- [4] Macronix, *MX25U12835F*, 1.8 V flash, PM1728 v1.9. <https://www.macronix.com/Lists/Datasheet/Attachments/8704/MX25U12835F,%201.8V,%20128Mb,%20v1.9.pdf>
- [5] Abracon, *ASE3 series*, 1.8 V oscillator supply/pin definitions. <https://abracon.com/Oscillators/ASE3series.pdf>
- [6] AMD, *7 Series PCB Design Guide*, UG483, capacitor placement p24 and current-path/plane inductance pp29–30. <https://docs.amd.com/api/khub/documents/6L8DUUei7ZCE78qwQafC3A/content>
- [7] AMD, *7 Series Packaging and Pinout*, UG475; package functions independently checked against the saved manufacturer xc7a100tcsg324pkg.csv. https://docs.amd.com/v/u/en-US/ug475_7Series_Pkg_Pinout
- [8] Samtec, QSH suggested footprint; final mating-height and assembly checks remain open. <https://sudden.docs.samtec.com/prints/qsh-xxx-01-x-d-xx-footprint.pdf>
- [9] Opal Kelly, XEM8310 powering documentation. It describes module input requirements, not a guaranteed exported supply for this headboard. <https://docs.opalkelly.com/xem8310/powering-the-xem8310/>

F.3 Current project evidence

Routing_Snapshot.json, All_Pin_Connections.csv, Component_List.csv, Net_Endpoints.csv, Final_Independent_Audit.json, Final_Rail_Invariant_Check.json, Final_Power_Review.md and the accompanying source/readback records.

The 117-signal names originate from the supplied team discussion and retained proposal, including the correction adding four SPI clocks. Those names do not close ASIC voltage/timing limits. Gerald's slides remain supporting context; the ASIC pad specification and owner confirmation are still required.